CSM 357

Human-Computer Interaction

lecture 3: Interaction models in Human-Computer Interaction

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In this section, we will consider the communication between user and system: the ***interaction***.

We will look at some **models of interaction** that enable us to identify and evaluate components of the interaction, and at the physical, social and organizational issues

Models

the usefulness of **models** to help us to understand complex behavior and complex systems.

What is interaction?

communication

user ↔ system

Interaction

- Interaction involves at least two participants: **the user and the system**.

- Both are **complex**, and are very different from each other in the way that they communicate and view the domain and the task.

- **The interface must therefore effectively translate between them to allow the interaction to be successful.**

Interaction

This translation can fail at a number of points and for a number of reasons.

The use of models of interaction can:

1. **help us to understand exactly what is going** on in the interaction and identify the likely root of difficulties.

2. They also **provide us with a framework to compare different interaction styles** and to consider interaction problems.

   eg. of interaction styles: command line, menus, form filling, question and answer

Some terms of interaction

- **domain** – an **area of expertise** and knowledge in some real-world activity e.g. graphic design
- **goal** – what you want to achieve
  e.g. create a solid red triangle
- **task** – how you go about doing it
  ultimately in terms of operations or actions
  e.g. … select fill tool, click over triangle

Interaction models

We begin by considering the most influential model of interaction, **Norman's *execution–evaluation cycle***; then we look at another model which extends the ideas of Norman's cycle.

Both of these models describe the interaction in terms of the **goals and actions of the user**

Traditionally, the purpose of an interactive system is to aid a user in accomplishing ***goals*** from some application ***domain***.

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The execution–evaluation cycle

Norman's model of interaction is perhaps the most influential in Human–Computer Interaction, possibly because of its closeness to our intuitive understanding of the interaction between human user and computer.

1. The user formulates a plan of action, which is then executed at the computer interface.

2. When the plan, or part of the plan, has been executed, the user observes the computer interface to evaluate the result of the executed plan, and to determine further actions.

Norman's model of interaction: The execution–evaluation cycle

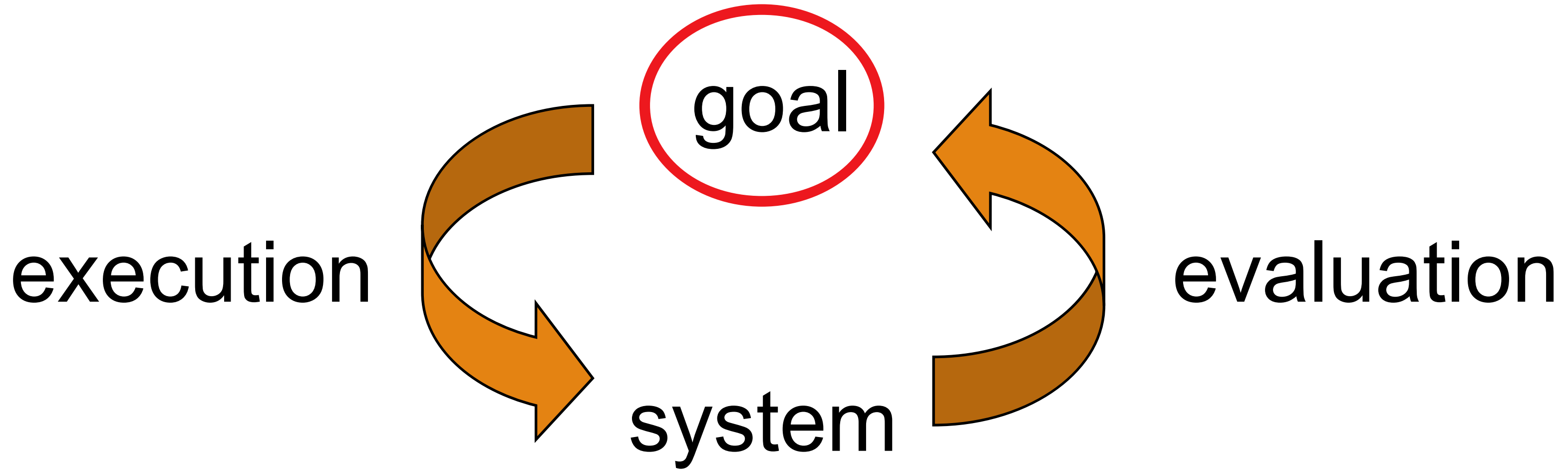
The Norman's model interactive cycle can be divided into **two major phases: execution and evaluation**. These can then be subdivided into further **stages**, **seven in all.**

The stages in Norman's model of interaction are as follows:

Stages in Norman’s model of interaction

1. Establishing the goal.
2. Forming the intention.
3. Specifying the action sequence.
4. Executing the action.
5. Perceiving the system state.
6. Interpreting the system state.
7. Evaluating the system state with respect to the goals and intentions.

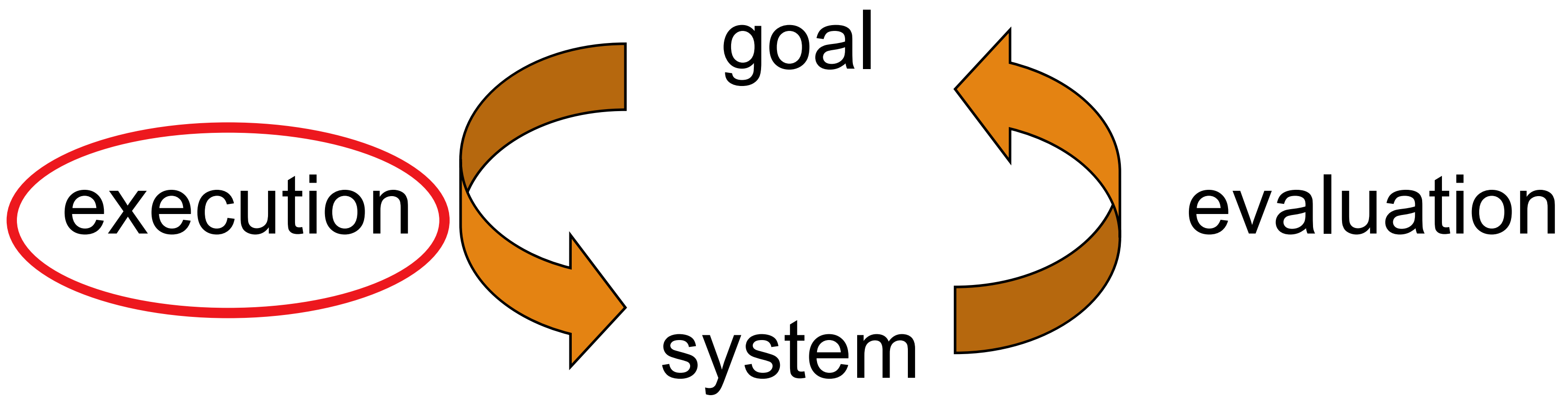
Stages in Norman's model of interaction (Simplified)



Each stage is an activity of the user.

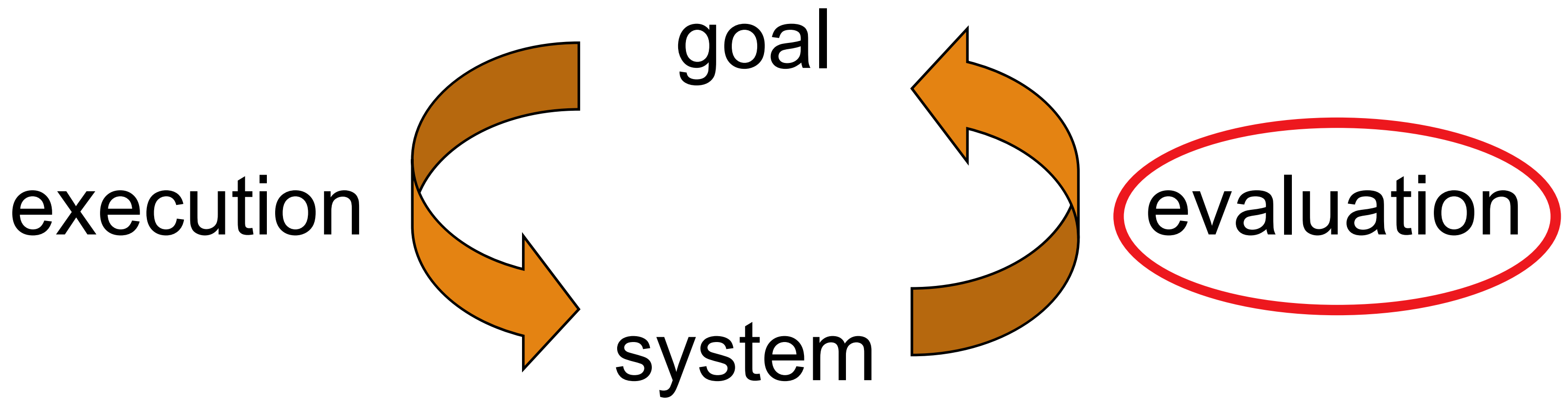
First the user forms a goal. This is the user's notion of what needs to be done and is framed in terms of the domain, in the task language.

Stages in Norman's model of interaction (Simplified)



It is liable to be imprecise and therefore needs to be **translated into the more specific intention**, and **the actual actions** that will reach the goal, before it can be executed by the user.

Stages in Norman's model of interaction (Simplified)



The user **perceives** the new state of the system, after **execution** of the action sequence, and **interprets** it in terms of his expectations.

If the system state reflects the user's goal then the computer has done what he wanted and the interaction has been successful; otherwise the user must formulate a new goal and repeat the cycle.

Practical example: Norman's model of interaction

Norman uses a simple example of switching on a light to illustrate this cycle.

1. Imagine you are sitting reading as evening falls. You decide you need more light; that is you **establish the goal** to get more light.
2. From there you **form an intention** to switch on the desk lamp,
3. and you **specify the actions required**, to reach over and press the lamp switch. If someone else is closer, the intention may be different. You may ask them to switch on the light for you. Your goal is the same but the intention and actions are different.
4. When you have **executed the action**
5. you **perceive the result**, either the light is on or it isn't

Practical example: Norman's model of interaction

example of switching on a light cont'd

6. **you interpret** the results, based on your knowledge of the world.

For example, if the light does not come on you may interpret this as indicating the bulb has blown or the lamp is not plugged into the mains, and you will **formulate new goals to deal with this**.

7. If the light does come on, **you will evaluate the new state according to the original goals**. Is there now enough light?

If so, the cycle is complete. If not, you may **formulate a new intention** to switch on the main ceiling light as well.

Why some interfaces cause problems to their users

Norman uses this model of interaction to demonstrate why some interfaces cause problems to their users.

He describes these in **terms of the *gulfs of execution* and the *gulfs of evaluation*.**

- the user and the system do not use the same terms to describe the domain and goals
- (we called the language of the system **the *core language*** and the language of the user the ***task language***.)

Gulf of Execution

The **gulf of execution** is the difference between the user’s formulation of the actions to reach the goal and the actions allowed by the system.

If the actions allowed by the system correspond to those intended by the user, the interaction will be effective.

The interface should therefore **aim to reduce this gulf.**

For example, a person can look at a light switch and easily tell what the current state of the system is (i.e., whether the light is on or off) and how to operate the switch.

This means that the gulf of execution is small. Norman states that, in order to design the best interfaces, the gulf must be kept as small as possible.

Gulf of Evaluation

Gulf of evaluation is the degree of ease with which a user can perceive and interpret whether or not the action they performed was successful.

This gulf is small when the system provides information about its state in a form that is easy to receive, interpret, and matches the way the person thinks of the system.

An example of a large gulf of evaluation is when an application has a spinning wheel to show a "loading" state after the user performs an action. The wheel alone is not enough for the user to interpret the progress that the system is making in response to their action. The gulf can be shortened by having a loading bar instead.

Gulf of Evaluation

The **gulf of evaluation** is the distance between the physical presentation of the system state and the expectation of the user.

If the user can readily evaluate the presentation in terms of his goal, the gulf of evaluation is small.

The more effort that is required on the part of the user to interpret the presentation, the less effective the interaction.

Individual activity

1. Pick an activity from your regular life that you perform with a computing device. For this activity, there should be some struggles with a large gulf of execution or gulf of evaluation, especially due to a weakness of the interface (software or hardware) involved in the activity.

First, describe what makes that gulf wide. What are the failures of the current interface to bridge the gulf?

2. **Then, select a similar activity from your regular life that does a better job bridging its gulf of execution or gulf of evaluation**.

Briefly describe that activity and what gives it a narrower gulf, then describe what lessons could be borrowed from the second activity to resolve the wide gulf in the first activity.

Individual activity

In selecting the activities, we recommend specifically selecting activities in similar domains. For example, you might select two appliances in your kitchen, or two mobile apps that do the same time

Adapted from http://omscs6750.gatech.edu/summer-2017/assignment-p1/

Human errors- Slips or Mistakes

Human error – slips and mistakes

Human errors are often classified into *slips* and *mistakes*. We can distinguish these using Norman's gulf of execution.

If you understand a system well you may know exactly what to do to satisfy your goals – you have formulated the correct action. However, perhaps you mistype or you accidentally press the mouse button at the wrong time. These are called *slips*; you have formulated the right action, but fail to execute that action correctly.

However, if you don't know the system well you may not even formulate the right goal. For example, you may think that the magnifying glass icon is the 'find' function, but in fact it is to magnify the text. This is called a *mistake*.

If we discover that an interface is leading to errors it is important to understand whether they are slips or mistakes. Slips may be corrected by, for instance, better screen design, perhaps putting more space between buttons. However, mistakes need users to have a better understanding of the systems, so will require far more radical redesign or improved training, perhaps a totally different metaphor for use.

Human errors- Slips or Mistakes

Slip

- understand system and goal
- correct formulation of action
- incorrect action

Mistake

- may not even have right goal!

Fixing things?

slip – better interface design

mistake – better understanding of system

Shortfalls of Norman’s model

Norman’s model is a useful means of understanding the interaction, in a way that is clear and intuitive. It allows other, more detailed, empirical and analytic work to be placed within a common framework.

However, it only considers the system as far as the interface. It concentrates wholly on the user’s view of the interaction.

It does not attempt to deal with the system’s communication through the interface.

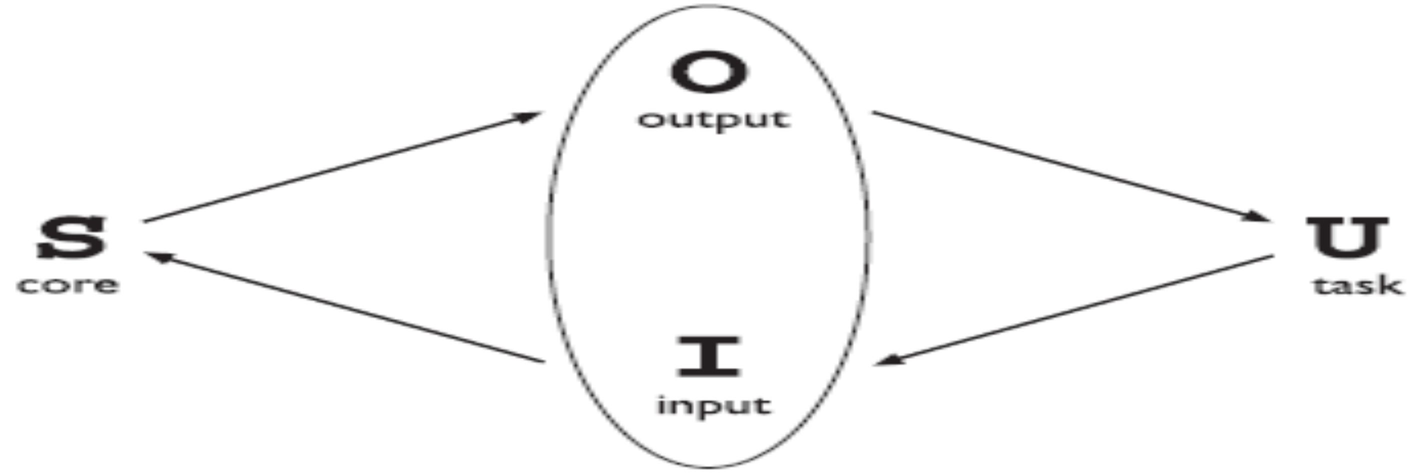
An extension of Norman’s model, **proposed by Abowd and Beale**, addresses this problem

Abowd and Beale's Interaction framework

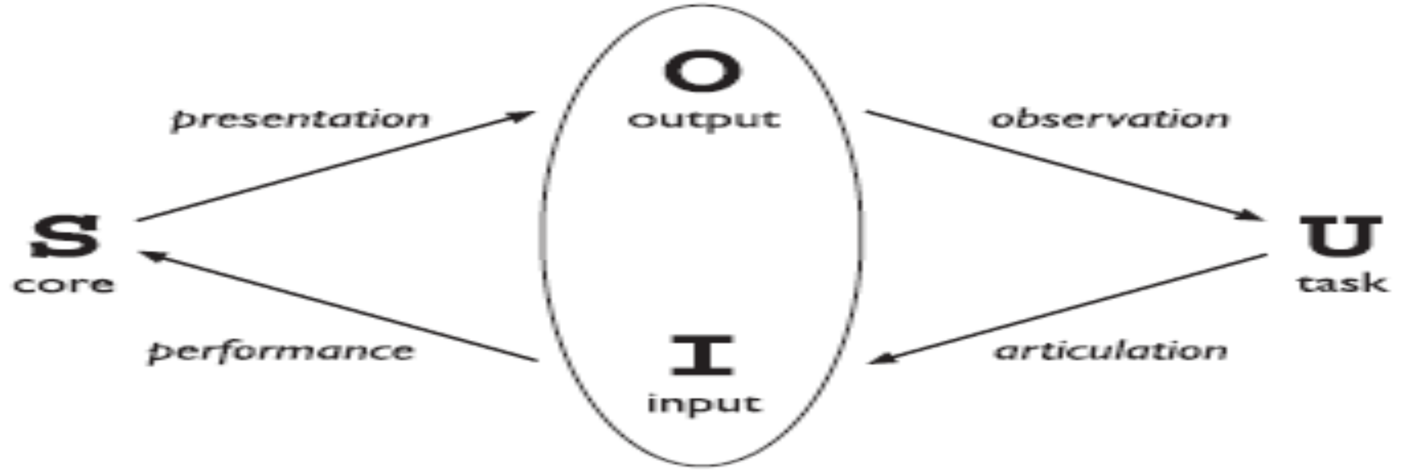
The interaction framework attempts a more realistic description of interaction by including the system explicitly, and **breaks it into four main components**, as shown in Figure 2.

The **nodes** represent the four major components in an interactive system: the ***System*****, the *****User*****, the *****Input***** and the *****Output***.

Each component has its own language.



The general interaction framework



Translations between components

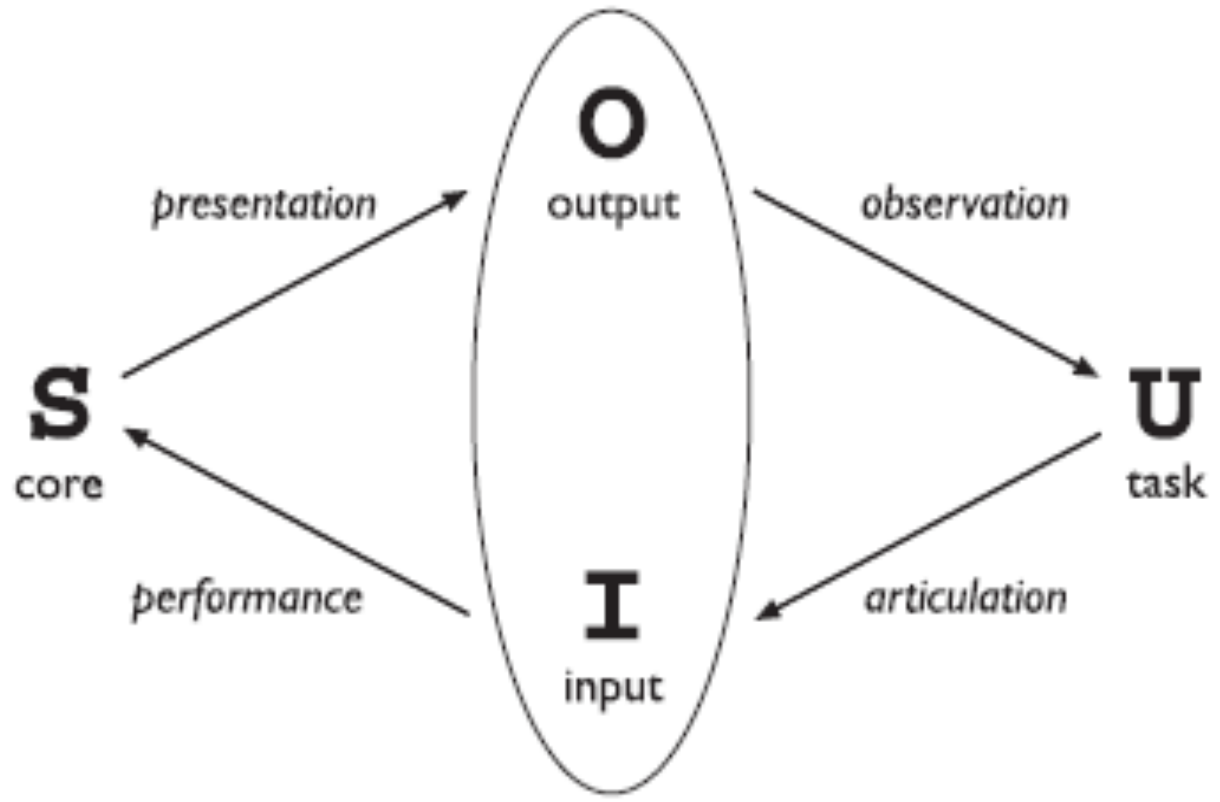
Figure 2

Abowd and Beale's Interaction framework

In addition to the ***User*'s task language** and the ***System*'s core language**, which we have already introduced, there are **languages for both the *Input* and *Output*** components.

***Input* and *Output* together form the *Interface*.**

As the interface sits between the *User* and the *System*, there are **four steps** in the interactive cycle, each corresponding to a translation from one component to another, as shown by the labeled arcs in the Figure on this slide

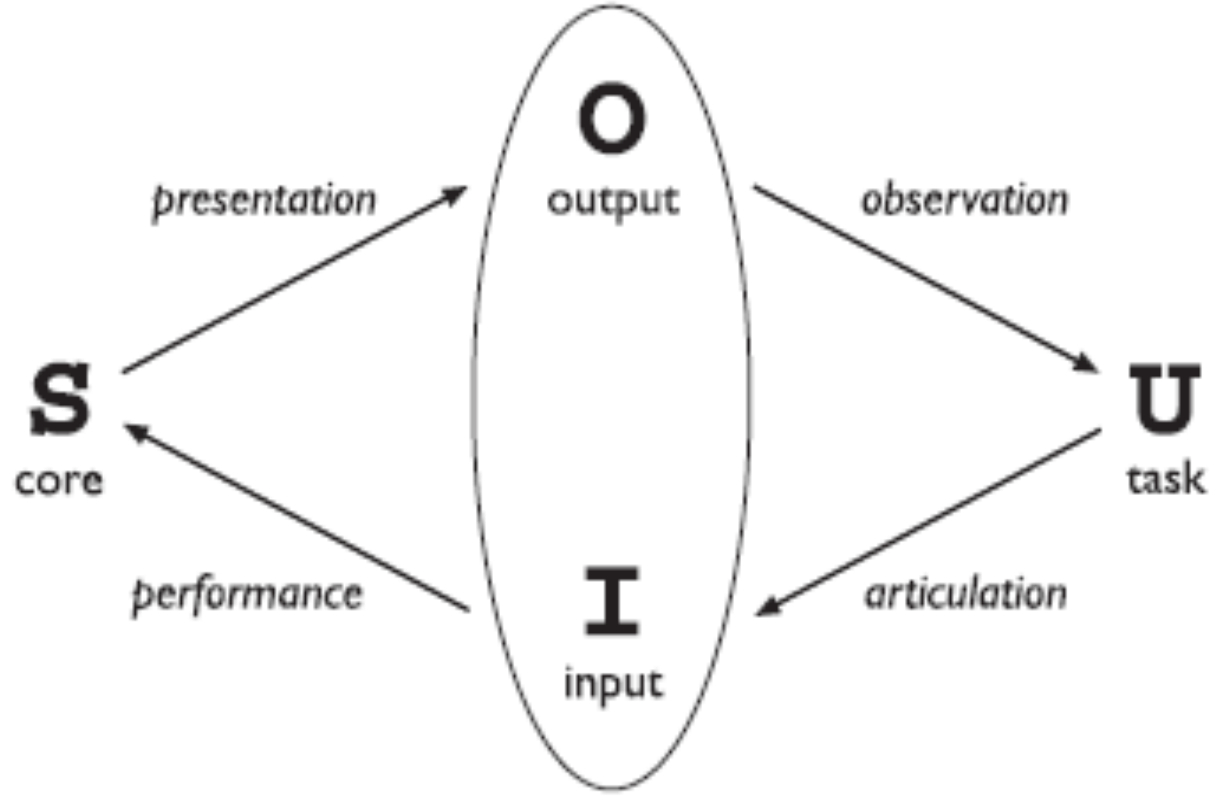


Translations between components

Abowd and Beale's Interaction framework

The *User* **begins** the interactive cycle with the **formulation of a goal** and a task to achieve that goal.

The **only way the user can manipulate the machine is through the *Input***, and so the task must be articulated within the **input language**.



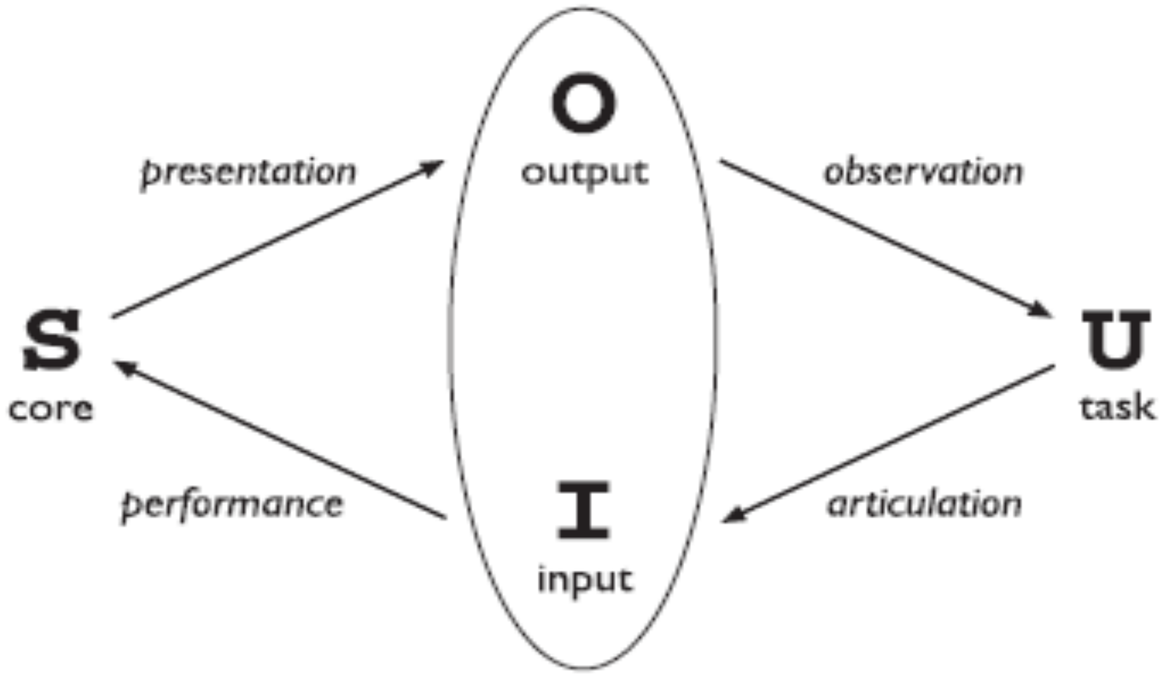
Translations between components

Abowd and Beale's Interaction framework

The input language is **translated** into the **core language** as operations to be performed by the *System*.

The *System* then transforms itself as described by the operations; the **execution phase** of the cycle is **complete** and the **evaluation** phase now **begins**.

The ***System* is in a new state**, which must now be **communicated** to the *User*.

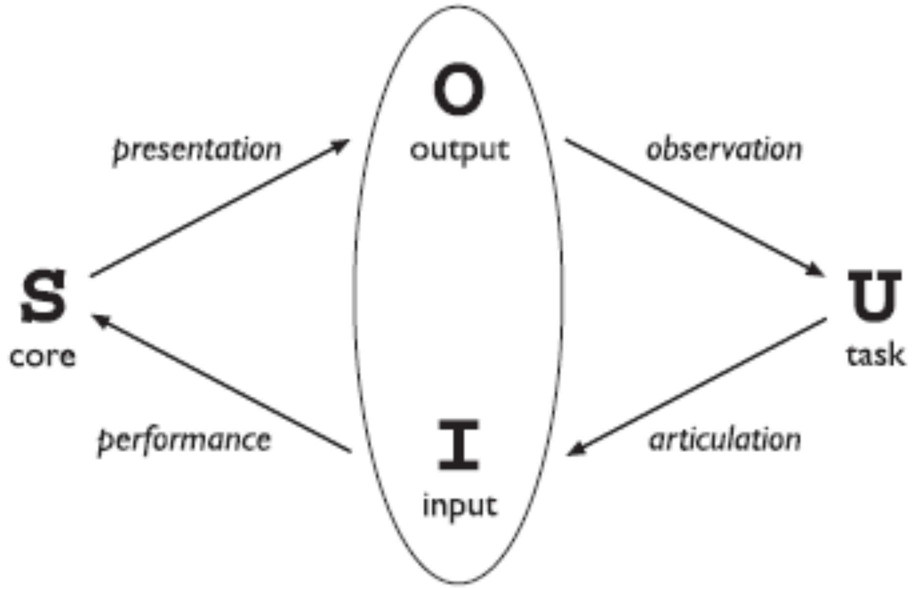


Translations between components

Abowd and Beale's Interaction framework

The current values of system attributes are rendered as concepts or features of the *Output*.

It is then up to the *User* to observe the *Output* and assess the results of the interaction relative to the original goal, ending the evaluation phase and, hence, the interactive cycle.



Translations between components

Frameworks and HCI

As well as providing a means of discussing the details of a particular interaction, **frameworks provide a basis for discussing other issues that relate to the interaction.**

The ACM SIGCHI (The Association for Computing Machinery *Special Interest Group on Computer-Human Interaction)* Curriculum Development Group presents a framework similar to that presented on earlier slides, and uses it to place different areas that relate to HCI

In Figure 3 on the next slide shows how different aspects relate to the interaction framework

A framework for human–computer interaction. Adapted from ACM SIGCHI Curriculum Development Group

the field of ***ergonomics*** addresses issues on the **user side of the interface**, covering both input and output, as well as the user's immediate context.

dialog is most usually **associated with the computer and so is biased to that side of the framework.**

Presentation and screen design relates to the **output** branch of the framework.

Each of these areas has important implications for the design of interactive systems and the performance of the user

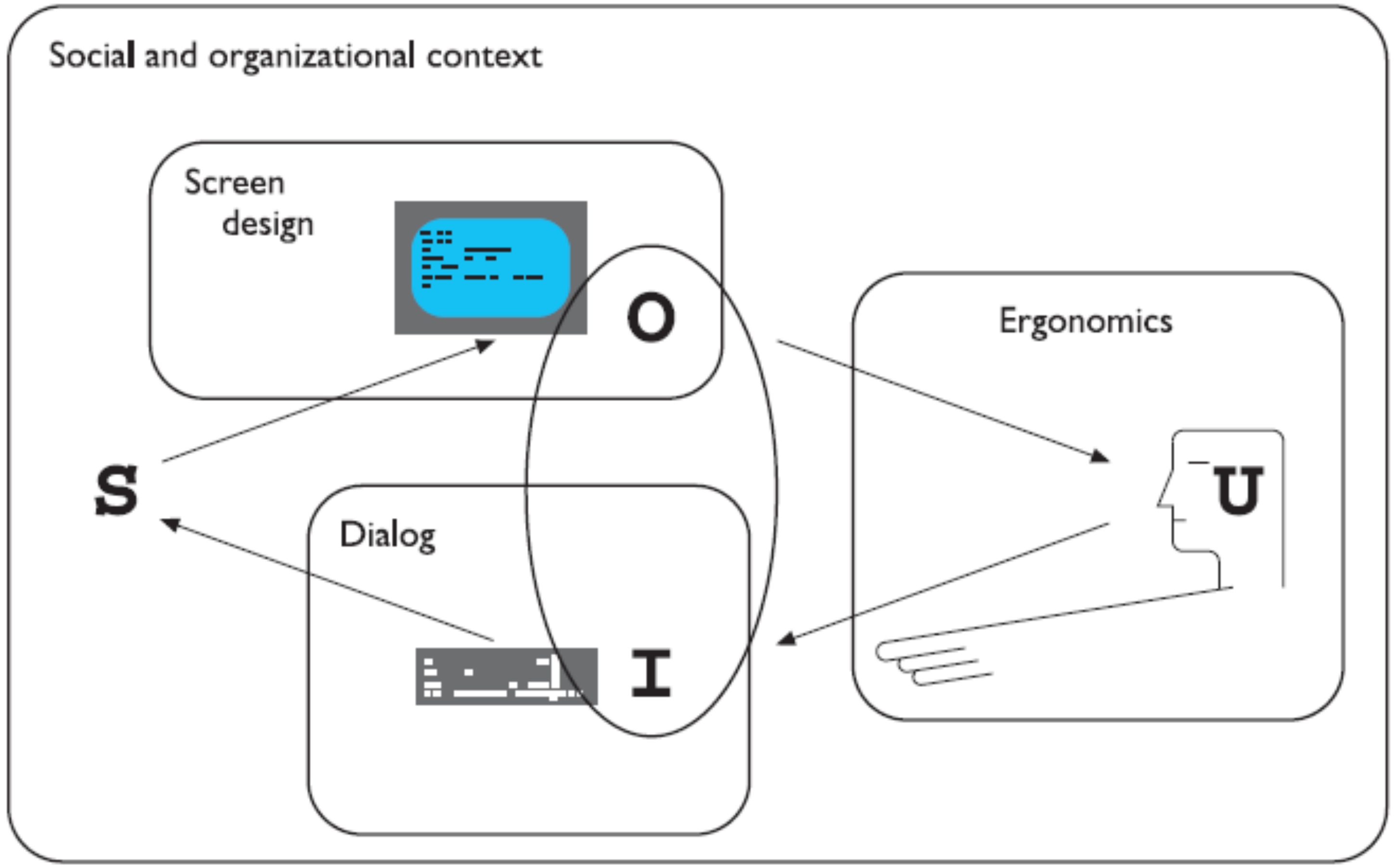


Figure 3.3 A framework for human–computer interaction. Adapted from ACM SIGCHI Curriculum Development Group [9]

Figure 3

ERGONOMICS

Ergonomics (or human factors) is traditionally the study of the physical characteristics of the interaction: how the controls are designed, the physical environment in which the interaction takes place, and the layout and physical qualities of the screen.

Ergonomics is a large and established field, which is closely related to but distinct from HCI, and full coverage would demand a course in its own right

However, let us consider some of the issues addressed by ergonomics relating to HCI

We will look at the arrangement of controls and displays, the physical environment, health issues and the use of color.

Ergonomics: Arrangement of controls and displays

- Sets of controls and parts of the display should be grouped logically to allow rapid access by the user
- **This may not seem so important when we are considering a single user of a spreadsheet on a PC, but it becomes vital when we turn to safety-critical applications such as plant control, aviation and air traffic control.**
- **In each of these contexts, users are under pressure and are faced with a huge range of displays and controls.**
- Therefore, it is crucial that the physical layout of these are appropriate.

Ergonomics: Arrangement of controls and displays

- Even for less critical PC application, inappropriate placement of controls and displays can lead to inefficiency and frustration.
- For example, on a particular electronic newsreader, the command key to **read** articles from a newsgroup (y) is **directly beside** the command key to **unsubscribe** from a newsgroup (u) on the keyboard.
- **This poor design frequently leads to inadvertent removal from newsgroups. Although this is recoverable** it wastes time and is annoying to the user.
- **We can therefore see that appropriate layout is important in all applications.**
- It is important to group controls together logically (**and keeping opposing controls separate**).

Ergonomics: Arrangement of controls and displays

The **exact arrangement** of controls will depend on the **domain** and the **application**, but possible arrangements include the following:

- **functional** controls and displays are organized so that those that are functionally related are placed together;
- **sequential** controls and displays are organized **to reflect the order of their use** in a typical interaction (this may be especially appropriate in domains where a particular task sequence is enforced, such as aviation);
- **frequency** controls and displays are organized according to how frequently they are used, with the most commonly used controls being the most easily accessible.

Ergonomics: Arrangement of controls and displays

In addition to the organization of the controls and displays in relation to each other, the **entire system interface must be arranged appropriately in relation to the user's position.**

So, for example, the user should be able to reach all controls necessary and view all displays without excessive body movement.

Critical displays should be at eye level.

Lighting should be arranged to avoid glare and reflection, distorting displays.

Controls should be spaced to provide adequate room for the user to manoeuvre.

Ergonomics: The physical environment of the interaction

As well as addressing physical issues in the layout and arrangement of the machine interface, **ergonomics is also concerned with the design of the work environment itself.**

Where will the system be used? By whom will it be used? Will users be sitting, standing or moving about?

Again, this will depend largely on the domain and will be more critical in specific control and operational settings than in general computer use.

However, the physical environment in which the system is used may influence how well it is accepted and even the health and safety of its users.

It should therefore be considered in all designs.

Ergonomics: The physical environment of the interaction

The first consideration here is the **size of the users**. Obviously this is going to vary considerably.

However, in any system the smallest user should be able to reach all the controls (this may include a user in a wheelchair), and the largest user should not be cramped in the environment.

In particular, all users should be comfortably able to see critical displays. For long periods of use, the user should be seated for comfort and stability.

Seating should provide back support.

If required to stand, the user should have room to move around in order to reach all the controls.

Ergonomics: The physical environment of the interaction

bear in mind the possible consequences of our designs on the health and safety of users.

obvious safety risks of poorly designed safety-critical systems(aircraft crashing, nuclear plant leaks and worse),

there are a number of factors that may affect the use of more general computers.

- **Physical position**
- **Temperature**
- **Lighting**
- **Noise**
- **Time**

Ergonomics: The use of colour

the **use of colour** in displays is an ergonomics issue.

Colours used in the display should be as distinct as possible and the distinction should not be affected by changes in contrast.

Blue should not be used to display critical information.

Ergonomics: The use of colour

The colors used should also correspond to common conventions and user expectations.

Red, green and yellow are colors frequently associated with stop, go and standby respectively.

- Therefore, red may be used to indicate emergency and alarms;
- green, normal activity; and
- yellow, standby and auxiliary function.

These conventions should not be violated without very good cause.

Ergonomics: The use of colour

However, colour conventions are culturally determined.

For example, red is associated with danger and warnings in most western cultures, but in China it symbolizes happiness and good fortune.

The colour of mourning is black in some cultures and white in others.

Awareness of the cultural associations of colour is particularly important in designing systems and websites for a global market.

Interaction styles

Interaction can be seen as a dialog between the computer and the user.

The **choice of interface style** can affect the interaction in positive or negative ways.

Interaction styles

There are a number interface styles including:

- command line interface
- menus
- natural language
- question/answer and query dialog
- form-fills and spreadsheets
- WIMP
- point and click
- three-dimensional interfaces.

Command Line Interface

The command line interface was the first interactive dialog style to be commonly used and, in spite of the availability of menu-driven interfaces, it is still widely used.

It provides a means of expressing instructions to the computer directly, using function keys, single characters, abbreviations or whole-word commands.

More commonly today, it is supplementary to menu-based interfaces, providing accelerated access to the system's functionality for experienced users.

Command Line Interface (Pros)

1. **Command line interfaces are powerful in that they offer direct access to system functionality** (as opposed to the hierarchical nature of menus)

2. They are also flexible: the command often has a number of options or parameters that will vary its behavior in some way, and it can be applied to many objects at once, making it useful for repetitive tasks

Read more on the advantages of command line interface

Command Line Interface (Cons)

1. Commands must be remembered, as no cue is provided in the command line to indicate which command is needed.

2. They are therefore better for expert users than for novices.

3. commands are sometimes obscure and vary across systems, causing confusion to the user and increasing the overhead of learning.

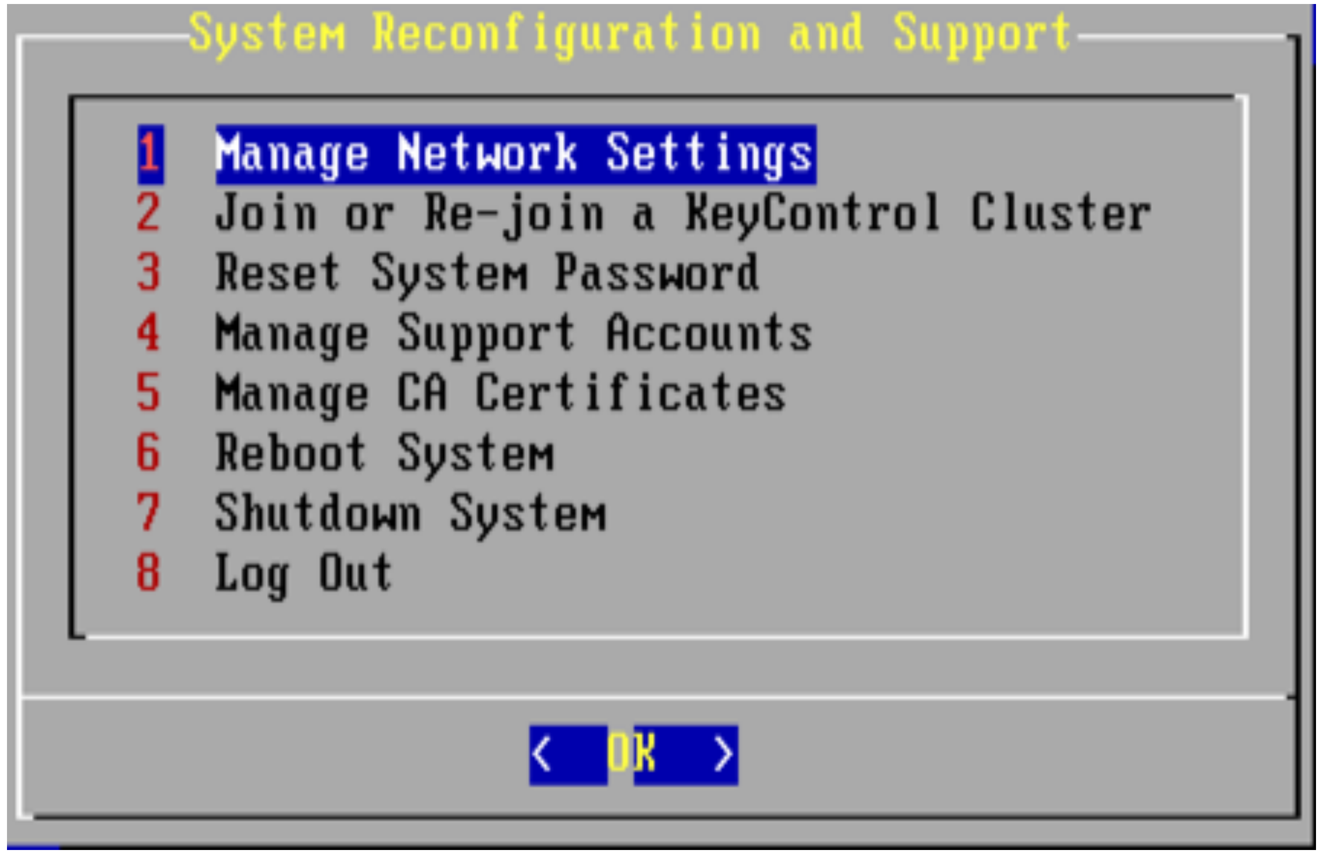
Read more on the disadvantages of command line interface

Menus

- **In a menu-driven interface, the set of options available to the user is displayed on the screen, and selected using the mouse, or numeric or alphabetic keys.**
- Since the options are visible **they are less demanding of the user, relying on recognition rather than recall.**
- However, **menu options still need to be meaningful and logically grouped** to aid recognition.
- naming of menu options cues for the user to find the required option.
- **Such systems either can be purely text based, or may have a graphical component**

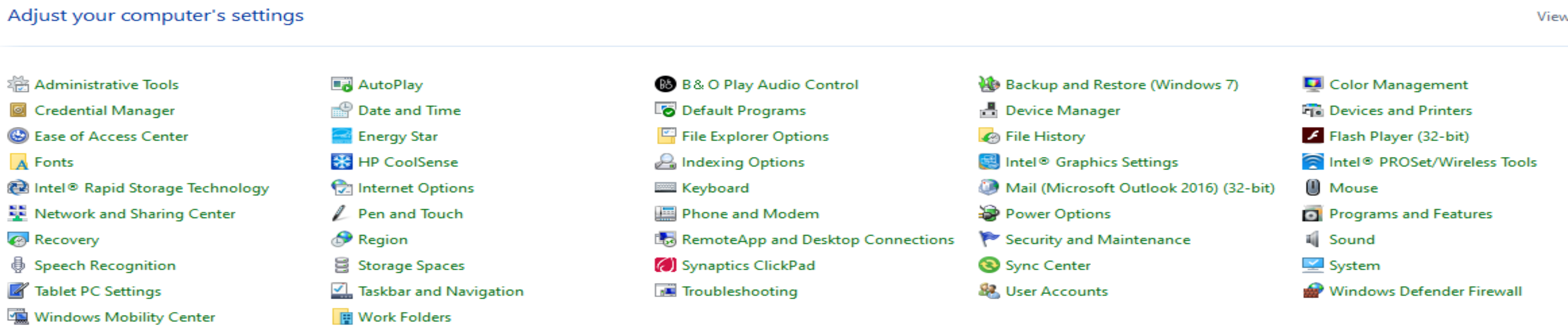
Text-based vs graphics-based menus

With **text-based menus**, with the menu options are presented as numbered choices



Example of text-based menu

With Graphics-based menus, menus appears within a rectangular box and choices are made, perhaps by typing the initial letter of the desired selection, or by entering the associated number, or by moving around the menu with the arrow keys or mouse.



Natural Language

Perhaps the most attractive means of communicating with computers, at least at first glance, is by natural language.

Users, unable to remember a command or lost in a hierarchy of menus, may long for the computer that is able to understand instructions expressed in everyday words!

Natural language understanding, both of speech and written input, is the subject of much interest and research currently.

Unfortunately, however, the ambiguity of natural language makes it very difficult for a machine to understand.

The ambiguity of natural language

Eg. Consider the sentence: The boy hit the dog with the stick

we cannot be sure whether the boy is using the stick to hit the dog or whether the dog is holding the stick when it is hit.

Even if a sentence's structure is clear, we may find ambiguity in the meaning of the words used.

For example, the word 'pitch' may refer to a sports field, a throw, a waterproofing substance or even, colloquially, a territory.

The ambiguity of natural language

- We often rely on the context and our general knowledge to sort out these ambiguities. This information is difficult to provide to the machine.
- Given these problems, it seems unlikely that a general natural language interface will be available for some time.
- However, systems can be built to understand restricted subsets of a language.
- For a known and constrained domain, the system can be provided with sufficient information to disambiguate terms.
- It is important in interfaces which use natural language in this restricted form that the user is aware of the limitations of the system and does not expect too much understanding.

Question/answer and query dialog

Question and answer dialog is a simple mechanism for providing input to an application in a specific domain.

The user is asked a series of questions (mainly with yes/no responses, multiple choice, or codes) and so is led through the interaction step by step.

An example of this would be web questionnaires.

These interfaces are easy to learn and use, but are limited in functionality and power.

As such, they are appropriate for restricted domains (particularly information systems) and for novice or casual users.

Question/answer and query dialog

Query languages, on the other hand, are used to construct queries to retrieve information from a database.

They use natural-language-style phrases, but in fact require specific syntax, as well as knowledge of the database structure.

Queries usually require the user to specify an attribute or attributes for which to search

the database, as well as the attributes of interest to be displayed.

This is straightforward where there is a single attribute, but becomes complex when multiple attributes are involved

Form-fills and spreadsheets

Form-filling interfaces are used primarily for data entry but can also be useful in data retrieval applications.

The user is presented with a display resembling a paper form, with slots to fill in.

Often the form display is based upon an actual form with which the user is familiar, which makes the interface easier to use.

The user works through the form, filling in appropriate values.

The data is then entered into the application in the correct place.

Form-fills and spreadsheets

Spreadsheets are a sophisticated variation of form filling.

The spreadsheet comprises a **grid of cells**, each of which can contain a **value** or a **formula**.

The formula can involve the values of other cells

The WIMP interface

Currently many common environments for interactive computing are examples of the *WIMP* interface style, often simply called windowing systems.

WIMP stands for windows, icons, menus and pointers (sometimes windows, icons, mice and pull-down menus), and is the default interface style for the majority of interactive computer systems in use today, especially in the PC and desktop workstation arena.

Examples of WIMP interfaces include Microsoft Windows, MacOS and many other software interfaces

Point-and-click interfaces

In most multimedia systems and in web browsers, virtually all actions take only a single click of the mouse button.

- You may point at a city on a map and when you click a window opens, showing you tourist information about the city.
- You may point at a word in some text and when you click you see a definition of the word. You may point at a recognizable iconic button and when you click some action is performed.

This point-and-click interface style is obviously closely related to the WIMP style.

Point-and-click interfaces

It clearly overlaps in the use of buttons, but may also include other WIMP elements.

However, the philosophy is simpler and more closely tied to ideas of *hypertext*.

In addition, the point-and-click style is not tied to mouse-based interfaces, and is also extensively used in touchscreen information systems.

The point-and-click style has been popularized by world wide web pages, which incorporate all the above types of point-and-click navigation: highlighted words, maps and iconic buttons.

Three-dimensional interfaces

There is an increasing use of three-dimensional effects in user interfaces.

The most obvious example is virtual reality, but VR is only part of a range of 3D techniques available to the interface designer.

The simplest technique is where ordinary WIMP elements, buttons, scroll bars, etc., are given a 3D appearance using shading, giving the appearance of being sculpted out of stone.

Summary

- Interaction models help us to understand what is going on in the interaction between user and system. They address the translations between what the user wants and what the system does.
- Ergonomics looks at the physical characteristics of the interaction and how these influence its effectiveness.
- The dialog between user and system is influenced by the style of the interface.
- The interaction takes place within a social and organizational context that affects both user and
- system.

Bank managers don't type . . .

The safe in most banks is operated by at least two keys, held by different employees of the bank. This makes it difficult for a bank robber to obtain both keys, and also protects the bank against light-fingered managers! ATMs contain a lot of cash and so need to be protected by similar measures. In one bank, which shall remain nameless, the ATM had an electronic locking device. The machine could not be opened to replenish or remove cash until a long key sequence had been entered. In order to preserve security, the bank gave half the sequence to one manager and half to another, so both managers had to be present in order to open the ATM. However, these were traditional bank managers who were not used to typing – that was a job for a secretary! So they each gave their part of the key sequence to a secretary to type in when they wanted to gain entry to the ATM. In fact, they both gave their respective parts of the key sequence to the *same* secretary. Happily the secretary was honest, but the moral is you cannot ignore social expectations and relationships when designing any sort of computer system, however simple it may be.

THANK YOU